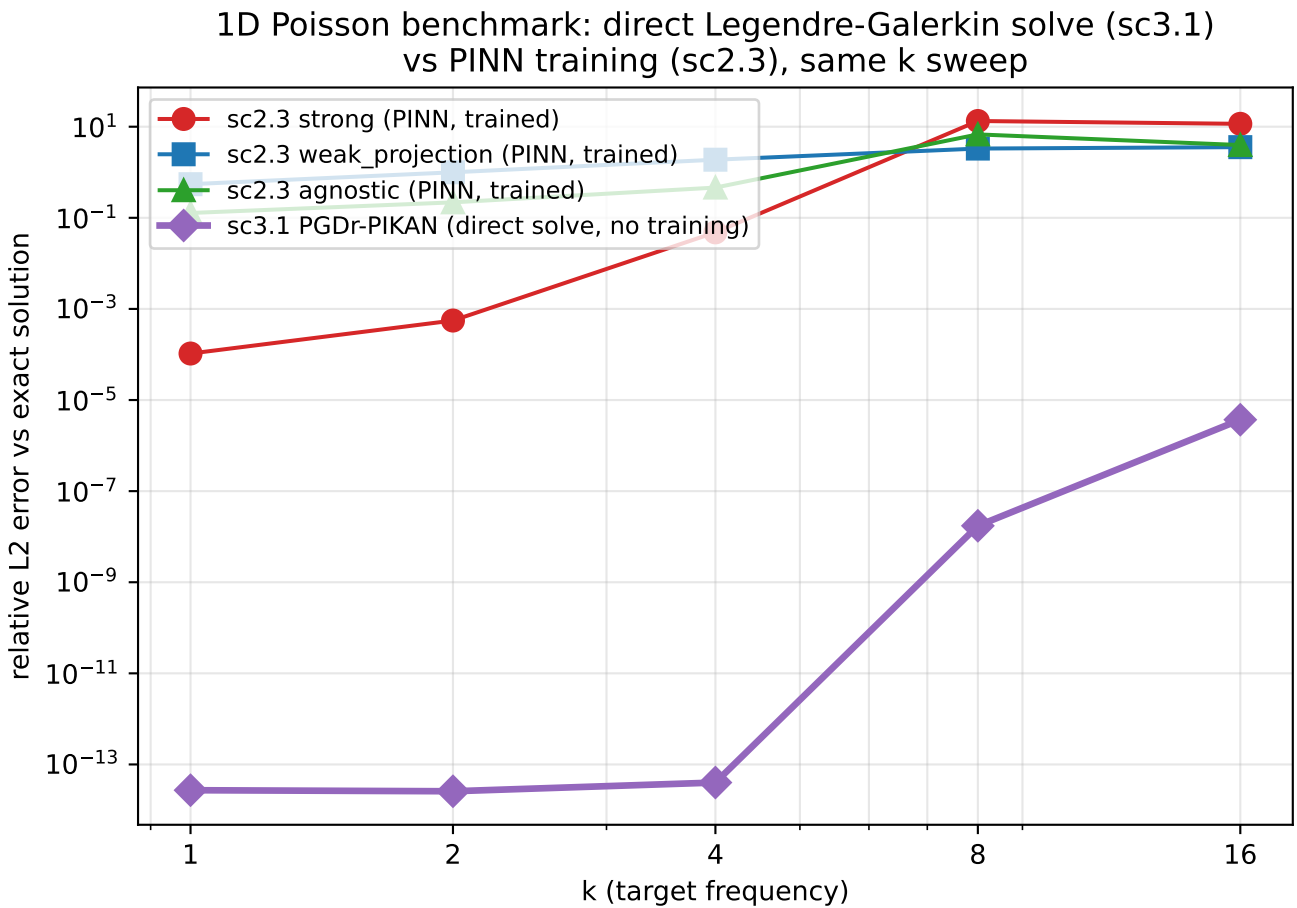
1D Poisson benchmark: direct Legendre-Galerkin solve (sc3.1)
vs PINN training (sc2.3), same k sweep
sc2.3 strong (PINN, trained)
sc2.3 weak_projection (PINN, trained)
sc2.3 agnostic (PINN, trained)
sc3.1 PGDr-PIKAN (direct solve, no training)
relative L2 error vs exact solution
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
1
2
4
8
16
k (target frequency)